



# Atomos Classes

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## RECAPITULATION TEST-1

Date: Sep 15, 2024

Time: 60 mins

Total Marks: 50 X 4 = 200

### Topics covered

#### Chemistry

**Some Basic Concepts Of Chemistry** - Nature of Matter, Properties of Matter and their Measurement, Uncertainty in Measurement, Laws of Chemical Combinations, Mole Concept and Molar Mass, Empirical Formula and Molecular Formula, Stoichiometry and its Calculations,

**Structure of Atom** - Particle Nature of Electromagnetic Radiation (EMR), Quantum Mechanical Model of Atom, Dual Behaviour of Matter - de Broglie Wave Equation, Quantum Numbers, Spectra, Heisenberg's Uncertainty Principle, Bohr's Model for Hydrogen Atom, Filling of Orbitals in Atom, Electronic Configuration of Elements and Magnetic Properties,

**Organic Chemistry: Some Basic Principles and Techniques** - Electron Displacement Effects or Polar Effects, Structural Isomerism,

Subject : Chemistry

No. of Questions : 50

- Statement A** : Solids have definite volume and shape.  
**Statement B** : In solids, the constituent particles are very close to each other and there is not much freedom of movement.

  - Statement A is correct but Statement B is incorrect.
  - Statement A is incorrect but Statement B is correct.
  - Both Statements are correct.
  - Both Statements are incorrect.
- Sulphur burns in oxygen to form sulphur dioxide. The properties of sulphur dioxide are**

  - Totally different from sulphur and oxygen
  - Similar to sulphur
  - Similar to oxygen
  - More similar to sulphur than oxygen
- What is the SI unit of density ?**

  - $\text{g cm}^{-3}$
  - $\text{gm}^{-3}$
  - $\text{kg m}^{-3}$
  - $\text{kg cm}^{-3}$
- Express 0.006006 into scientific notation in three significant digits:**

  - $6.01 \times 10^{-3}$
  - $6.006 \times 10^{-3}$
  - $6.00 \times 10^{-3}$
  - $6.0 \times 10^{-3}$
- Given the numbers: 161cm, 0.161 cm, 0.0161cm. The number of significant figures for the three numbers are**

  - 3, 4 and 5 respectively
  - 3, 3 and 3 respectively

(3) 3, 3 and 4 respectively

(4) 3, 4 and 4 respectively

- Assertion** : Atoms can neither be created nor destroyed.  
**Reason** : Under similar conditions of temperature and pressure, equal volume of gases does not contain equal number of molecules.

  - Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
  - Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
  - Assertion is correct but Reason is incorrect.
  - Assertion is incorrect but Reason is correct.
- Assertion** : Both 44 gCO<sub>2</sub>, and 16 gCH<sub>4</sub> have same number of carbon atoms.  
**Reason** : Both contain 1 g atom of carbon which contains  $6.023 \times 10^{23}$  carbon atoms.

  - Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
  - Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
  - Assertion is correct but Reason is incorrect.
  - Assertion is incorrect but Reason is correct.
- The following substances are present in different containers**

  - one gram atom of nitrogen
  - one mole of calcium
  - One atom of silver
  - One mole of oxygen molecules
  - $10^{23}$  atoms of carbon
  - One gram of iron

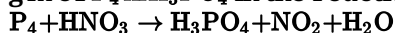
The correct order of increasing masses (in grams) is/are

- (1) (iii) < (iv) < (i) < (v)
- (2) (iii) < (vi) < (iv) < (ii)
- (3) (vi) < (v) < (i) < (iv)
- (4) (2) & (3) both are correct

9. Empirical formula of a compound is CH. If the molar mass of the compound is 78, the molecular formula is

- (1) C<sub>6</sub>H<sub>6</sub>
- (2) C<sub>3</sub>H<sub>3</sub>
- (3) C<sub>2</sub>H<sub>2</sub>
- (4) C<sub>3</sub>H<sub>6</sub>

10. What weight of HNO<sub>3</sub> is needed to convert 62 gm of P<sub>4</sub> in H<sub>3</sub>PO<sub>4</sub> in the reaction?



- (1) 63 gm
- (2) 630 gm
- (3) 315 gm
- (4) 126 gm

11. A bulb emits electromagnetic radiation of 660 nm wavelength. The total energy of radiation is  $3 \times 10^{-18}$  J. The number of emitted photons will be

$$(h = 6.6 \times 10^{-34} \text{ J s}, c = 3 \times 10^8 \text{ m/s})$$

- (1) 1
- (2) 10
- (3) 100
- (4) 1000

12. Total number of electrons that can be accommodated in the sub-shell (l = 2) are

- (1) 6
- (2) 10
- (3) 14
- (4) 2

13. The de - Broglie wavelength associated with a ball of mass 1 kg having kinetic energy 0.5 J, is

- (1)  $6.626 \times 10^{-34}$  m
- (2)  $13.20 \times 10^{-34}$  m
- (3)  $10.38 \times 10^{-21}$  m
- (4)  $6.626 \times 10^{-34}$  A

14. Identify an orbital with the quantum numbers n = 4, l = 3, m = 0.

- (1) 4f
- (2) 4p
- (3) 4s
- (4) 4d

15. Assuming that a 25 watt bulb emits monochromatic yellow light of wavelength  $0.57 \times 10^{-6}$  m. The rate of emission of quanta per second will be

- (1)  $5.89 \times 10^{15} \text{ sec}^{-1}$
- (2)  $7.28 \times 10^{17} \text{ sec}^{-1}$
- (3)  $5 \times 10^{10} \text{ sec}^{-1}$
- (4)  $7.18 \times 10^{19} \text{ sec}^{-1}$

16. Last line of Lyman series for H-atom has wavelength  $\lambda_1$  Å. The 2<sup>nd</sup> line of Balmer series has wavelength  $\lambda_2$  Å then

- (1)  $\frac{16}{\lambda_1} = \frac{9}{\lambda_2}$
- (2)  $\frac{16}{\lambda_2} = \frac{3}{\lambda_1}$
- (3)  $\frac{4}{\lambda_1} = \frac{1}{\lambda_2}$
- (4)  $\frac{16}{\lambda_1} = \frac{3}{\lambda_2}$

17. The Heisenberg uncertainty principle can be applied to

- (1) A cricket ball
- (2) A foot ball
- (3) A jet aeroplane
- (4) An electron

18. Energy of an electron is given by

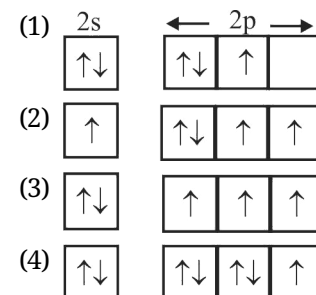
$$E = -2.178 \times 10^{-18} \text{ J} \left( \frac{Z^2}{n^2} \right). \text{ Wavelength of light}$$

required to excite an electron in an hydrogen atom from level n = 1 to n = 2 will be

$$(h = 6.62 \times 10^{-34} \text{ J.s and } c = 3.0 \times 10^8 \text{ m/s})$$

- (1)  $1.214 \times 10^{-7}$  m
- (2)  $2.816 \times 10^{-17}$  m
- (3)  $6.500 \times 10^{-17}$  m
- (4)  $8.500 \times 10^{-7}$  m

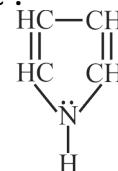
19. The orbital diagram in which 'Aufbau principle' is violated is



20. d<sup>6</sup> configuration will result in total spin of

- (1)  $\frac{3}{2}$
- (2)  $\frac{1}{2}$
- (3) 2
- (4) 1

21. How many delocalised  $\pi$ -electrons are there in following molecule?



- (1) 2
- (2) 4
- (3) 6
- (4) 8

22. The carboxylic acids and esters are

- (1) Position isomers
- (2) Functional isomers
- (3) Metamers
- (4) Chain isomers

23. Assertion: Carbon-oxygen bonds are of equal length in carbonate ion.

Reason: Bond length decreases with the multiplicity of bond between two atoms.

- (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
- (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
- (3) Assertion is correct but Reason is incorrect.
- (4) Assertion is incorrect but Reason is correct.

24. Identify the functional group that has electron donating inductive effect

- (1) -COOH
- (2) -CN
- (3) -CH<sub>3</sub>
- (4) -NO<sub>2</sub>

25. Inductive effect of which atom is taken as zero to compare inductive effect of other atoms?

- (1) Hydrogen
- (2) Chlorine
- (3) Carbon
- (4) Oxygen

## 26. Inductive effect involves the

- (1) Displacement of  $\sigma$  - electrons
- (2) Delocalization of  $\pi$  - electrons
- (3) Delocalization of  $\sigma$  - electrons
- (4) Displacement of  $\pi$  - electrons

## 27. Resonance is due to

- (1) Delocalisation of sigma electrons
- (2) Migration of H atoms
- (3) Migration of protons
- (4) Delocalisation of  $\pi$  electrons

## 28. In which of the following compounds the carbon marked with asterisk is expected to have greatest positive charge?

- (1)  $^*\text{CH}_3 - \text{CH}_2 - \text{Cl}$
- (2)  $^*\text{CH}_3 - \text{CH}_2 - \text{Mg}^+\text{Cl}^-$
- (3)  $^*\text{CH}_3 - \text{CH}_2 - \text{Br}$
- (4)  $^*\text{CH}_3 - \text{CH}_2 - \text{CH}_3$

## 29. Consider the following statements.

- I. The resonance structure have different positions of nuclei.
- II. Resonance structures have the same number of unpaired electrons.
- III. Among the resonance structures, the one which has more number of covalent bonds, all the atoms with octet of electrons and more dispersal of charge is more stable than others.

Read the given statements and choose the incorrect option.

- (1) Only I
- (2) Both II and III
- (3) Only III
- (4) All the statements are incorrect.

## 30. Mesomeric effect involve delocalisation of

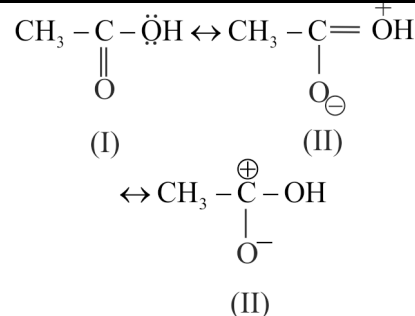
- (1)  $\pi$ -electrons
- (2)  $\sigma$ -electrons
- (3) Protons
- (4) Neutrons

 31. **Assertion:** Cycloheptatrienyl cation is aromatic.

**Reason :** Aromatic molecules have high degree of thermodynamic stability.

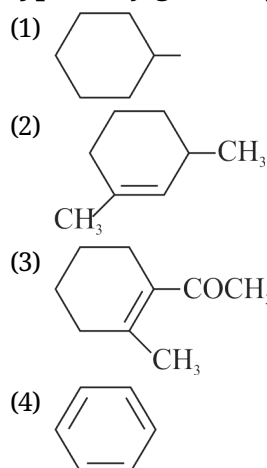
- (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
- (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
- (3) Assertion is correct but Reason is incorrect.
- (4) Assertion is incorrect but Reason is correct.

## 32. Compare the stability of resonating structures:



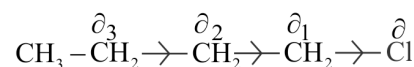
- (1) I > II > III
- (2) III > II > I
- (3) I > III > II
- (4) II > I > III

## 33. In which of the following molecules all the effects namely inductive, mesomeric, and hyperconjugation operate?



## 34. The C – C bond length of the following molecules is in the order

- (1)  $\text{C}_2\text{H}_6 > \text{C}_2\text{H}_4 > \text{C}_6\text{H}_6 > \text{C}_2\text{H}_2$
- (2)  $\text{C}_2\text{H}_2 < \text{C}_2\text{H}_4 < \text{C}_6\text{H}_6 < \text{C}_2\text{H}_6$
- (3)  $\text{C}_2\text{H}_6 > \text{C}_2\text{H}_2 > \text{C}_6\text{H}_6 > \text{C}_2\text{H}_4$
- (4)  $\text{C}_2\text{H}_4 > \text{C}_2\text{H}_6 > \text{C}_6\text{H}_2 > \text{C}_6\text{H}_6$

 35. Identify lowest positive charge developed (indicated by  $\delta$ ,  $\delta_1$ ,  $\delta_2$ ,  $\delta_3$ ) due to inductive effect in following compounds.


- (1)  $\delta_2$
- (2)  $\delta_3$
- (3)  $\delta$
- (4)  $\delta_1$

## 36. Which statement is correct for inductive effect?

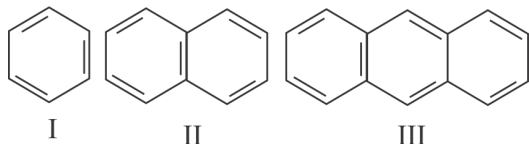
- (1) It is a permanent effect
- (2) It is the property of single bond
- (3) It causes permanent polarisation in the molecule
- (4) All are correct

## 37. Decreasing - I effect of given groups is

- (A)  $-\text{CN}$
- (B)  $-\text{NO}_2$
- (C)  $-\text{NH}_3^+$
- (D)  $-\text{F}$

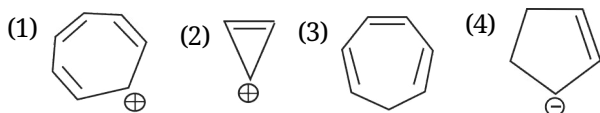
- (1) B > A > D > C
- (2) B > C > D > A
- (3) C > B > D > A
- (4) C > B > A > D

## 38. Find order of resonance energy of following molecules &amp; resonance energy/benzene also.



- (1) R.E  $I > II > III$   
R.E/benzene  $\Rightarrow I > II > III$
- (2) R.E  $\Rightarrow III > II > I$   
R.E/benzene  $\Rightarrow III > II > I$
- (3) R.E  $\Rightarrow I > II > III$   
R.E/benzene  $\Rightarrow III > II > I$
- (4) R.E R. E  $\Rightarrow III > II > I$   
R.E/benzene =  $I > II > III$

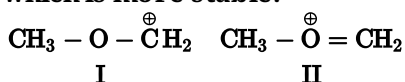
39. Which compound amongst the following is not an aromatic compound?



40. Identify -I effect causing group from the following

- (1)  $-\text{COOR}$  (2)  $-\text{CH}_3$   
(3)  $-\text{C}_2\text{H}_5$  (4)  $-\text{C}_3\text{H}_7$

41. The possible resonance structures for  $\text{CH}_3 - \ddot{\text{O}} - \overset{+}{\text{C}}\text{H}_2$  are given. Predict the structure which is more stable.



- (1) I only (2) II only  
(3) Both I and II (4) Can't be predicted

42. Identify the species from following that exhibits no bond resonance.

- (1)  $\text{CH}_3\text{CH}_2\text{Br}$  (2)  $\text{CH}_3\text{CH}_2(+)$   
(3)  $\text{CH}_3\text{CH}_2\text{NO}_2$  (4)  $\text{C}_6\text{H}_6$

43. In which of the following, resonance will be possible?

- (1)  $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CHO}$   
(2)  $\text{CH}_2 = \text{CH} - \text{CH} = \text{O}$   
(3)  $\text{CH}_3\text{COCH}_3$   
(4)  $\text{CH}_2 = \text{CH} - \text{CH}_2 - \text{CH} = \text{CH}_2$

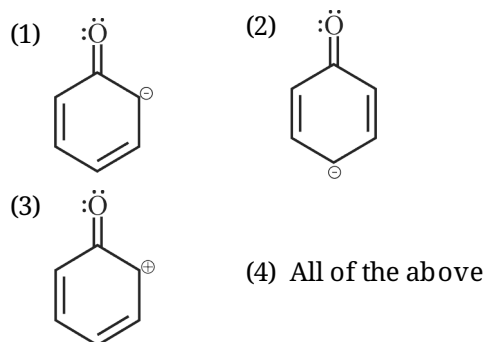
44. **Statement A :** Inductive effect and resonance effect cause permanent polarisation of bond.

**Statement B :** Inductive effect involve electron displacements due to the influence of an atom or a substituted group present in the molecule.

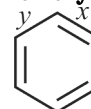
- (1) Statement A is correct but Statement B is incorrect.  
(2) Statement A is incorrect but Statement B is correct.

- (3) Both statements are correct.  
(4) Both Statements are incorrect.

45. Which is not the resonance structure of phenoxide ion?

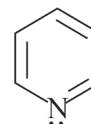


46. C - C bond-lengths x and y in benzene are



- (1)  $x > y$  (2)  $x < y$   
(3)  $x = y$  (4) Data insufficient

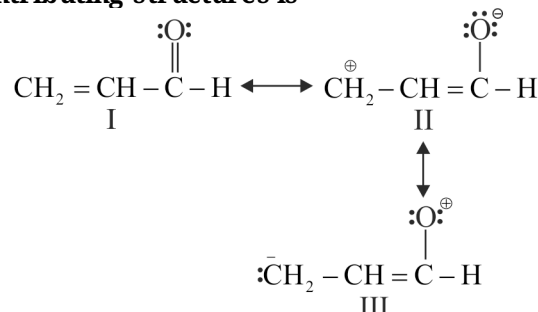
47. In pyridine;



Number of conjugated electrons are

- (1) 6 (2) 8 (3) Zero (4) 5

48. The order of relative stability of the contributing structures is



Choose the correct answer from the options given below :

- (1)  $I = II = III$  (2)  $II > I > III$   
(3)  $I > II > III$  (4)  $III > II > I$

49. Which of the following groups has the highest +I effect?

- (1)  $\text{CH}_3 -$  (2)  $\text{CH}_3$   
(3)  $(\text{CH}_3)_2\text{CH} -$  (4)  $(\text{CH}_3)_3\text{C} -$

50. Which of the following ions does not show resonance?

- (1)  $\text{NO}_3^-$  (2)  $\text{CH}_3\text{COO}^-$   
(3)  $\text{Cl}^-$  (4)  $\text{CO}_3^{2-}$